

Problem 1) One in which the motion comes back to each position it has visited after a specified time with the same velocity.

Problem 2) ① Simple harmonic, damped, driven-damped.

② $\ddot{x}$: acceleration ; $2\beta\dot{x}$: dissipative force $\omega_0^2 x$: restoring force

③ The factor 2 simplifies the final solution to the ODE and has no impact on the solution.

Problem 3) The tangential component of $-mg$ is,
 $-mg \sin \theta = m(l\ddot{\theta})$

This eqn. is not analytically solvable.

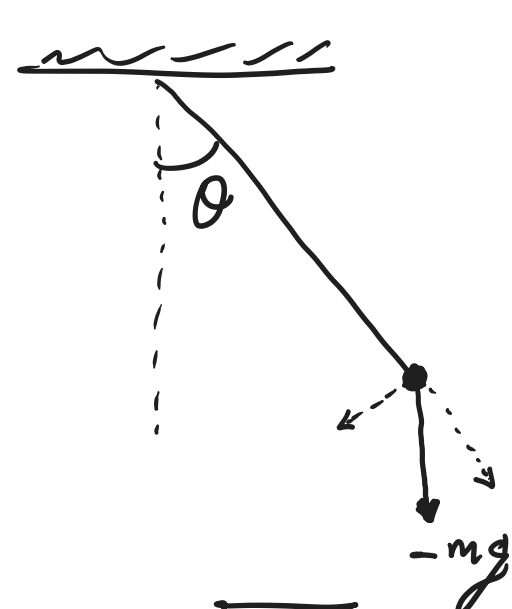
We must make a small angle approximation.

$$\lim_{\theta \rightarrow 0} \sin \theta = \theta$$

$$\Rightarrow \ddot{\theta} + \omega^2 \theta \cong 0 ; \theta \rightarrow 0 \text{ where } \omega = \sqrt{\frac{g}{l}}$$

$$\Rightarrow \theta(t) \cong A \cos(\omega t + \phi) \text{ for small } \theta.$$

A and ϕ are unknown constants determined by the initial conditions of the problem.



Problem 4) ① A is the amplitude.

$$\textcircled{2} \quad x(t) = A \sin(\omega t - \delta) \Rightarrow \dot{x}(t) = A\omega \cos(\omega t - \delta)$$

$$\Rightarrow \ddot{x}(t) = -A\omega^2 \sin(\omega t - \delta)$$

$$\Rightarrow \boxed{\ddot{x} + \omega^2 x = 0} \text{ simple harmonic motion.}$$

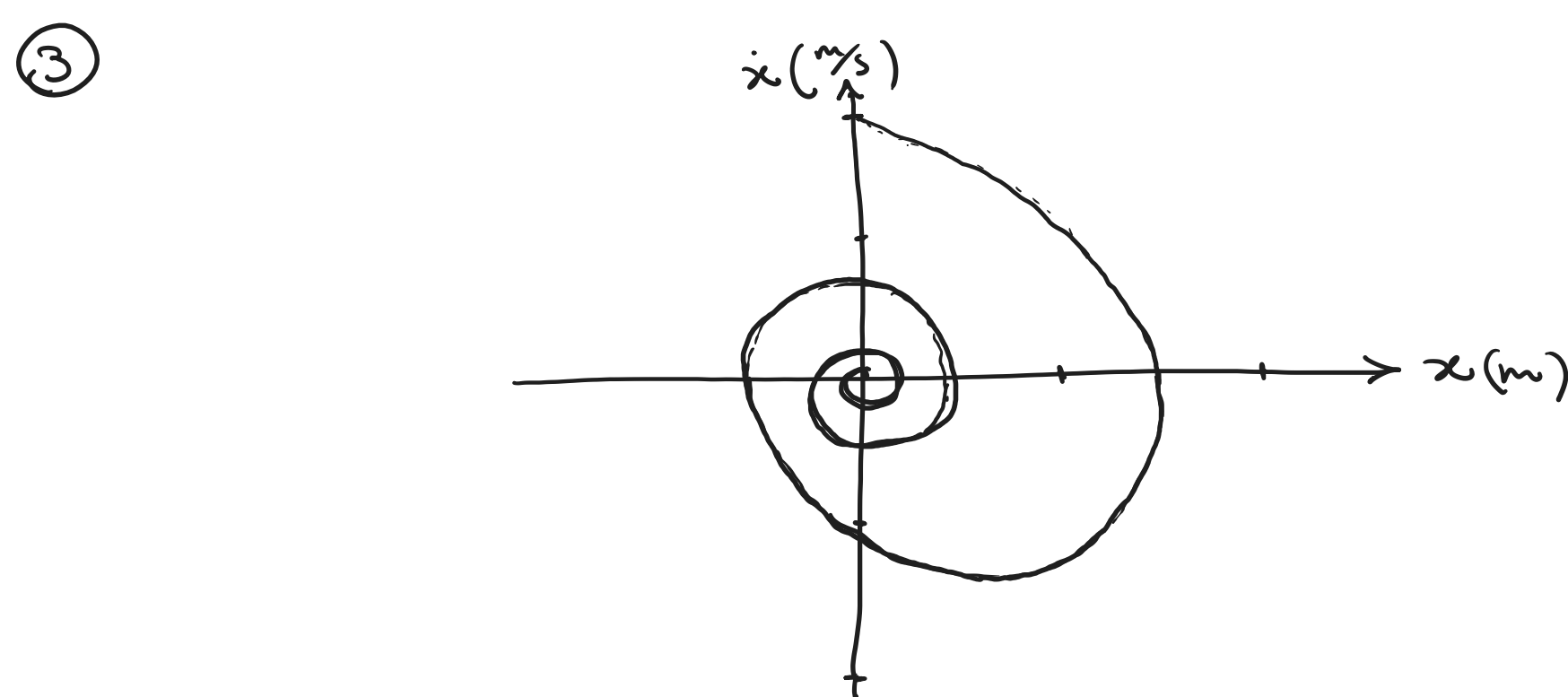
$$\textcircled{3} \quad \left. \begin{array}{l} x = A \sin(\omega t - \delta) \\ \dot{x} = A\omega \cos(\omega t - \delta) \end{array} \right\} \Rightarrow \left(\frac{x}{A} \right)^2 + \left(\frac{\dot{x}}{A\omega} \right)^2 = \sin^2(\cdot) + \cos^2(\cdot) = 1$$

This is the eqn of an ellipse. ✓

Problem 5) ① Underdamped motion.

$$\textcircled{2} \quad x(t) = A e^{(-\frac{\beta}{2}t)} \sin(\omega t - \delta)$$

$$\Rightarrow \dot{x}(t) = -A e^{-\beta t} \left[\beta \cos(\omega t - \delta) + \omega \sin(\omega t - \delta) \right] ; T = \frac{1}{2} m \dot{x}^2$$



Problem 6) $L = T - U$; The kinetic energy is composed of radial and tangential components,

$$T = \frac{1}{2} m \left(\dot{x}^2 + ((l+x)\dot{\theta})^2 \right) \textcircled{I}$$

The potential energy has gravity and spring components;

$$U(x, \theta) = -mg(l+x) \cos \theta + \frac{1}{2} kx^2 \textcircled{II}$$

$$\textcircled{I} \textcircled{II} \Rightarrow L = \frac{1}{2} m \left(\dot{x}^2 + (l+x)^2 \dot{\theta}^2 \right) + mg(l+x) \cos \theta - \frac{1}{2} kx^2$$

There are two variables here: (x, θ)

$$\Rightarrow \left\{ \begin{array}{l} \frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} = 0 \Rightarrow m\ddot{x} = m(l+x)\dot{\theta}^2 + mg \cos \theta - kx \textcircled{III} \\ \frac{\partial L}{\partial \theta} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} = 0 \Rightarrow \frac{d}{dt} (m(l+x)^2 \dot{\theta}) = -mg(l+x) \sin \theta \textcircled{IV} \end{array} \right.$$

$$\Rightarrow m(l+x)^2 \ddot{\theta} + 2m(l+x)\dot{x}\dot{\theta} = -mg(l+x) \sin \theta$$

$$\Rightarrow m(l+x)\ddot{\theta} + 2m\dot{x}\dot{\theta} = -mg \sin \theta \textcircled{V}$$

III is simply the Newton's law $F=ma$ with the second term being related to the centripetal acceleration $-(l+x)\dot{\theta}^2$.

IV states that torque equals the rate of change of angular momentum.